Software Requirements Specification

for

Employee Management System

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the Employee Management System (EMS), a full-stack web application designed to digitize and streamline core HR operations including employee record management, attendance tracking, and leave management. This document is intended for the project supervisor, and any evaluators reviewing the project.

1.2 Document Conventions

Functional requirements are labeled using the format FR-<Module>-<Number> (e.g., FR-AUTH-01). Priority levels are marked as High, Medium, or Low.

1.3 Project Scope

The EMS aims to replace manual/spreadsheet-based employee record keeping with a centralized, role-based web application. The system will allow organizations to manage employee data, departmental structure, daily attendance,payroll processing and leave requests through a single platform. The initial release (v1) focuses on core HR workflows; payroll processing, performance reviews, and notification systems are identified as future scope beyond this release.

1.4 References

FastAPI Documentation — https://fastapi.tiangolo.com
React Documentation — https://react.dev
PostgreSQL Documentation — https://www.postgresql.org/docs

2. Overall Description

2.1 Product Perspective

EMS is a new, standalone full-stack application built using FastAPI (backend) and React (frontend), with a relational database for persistent storage. It is not a replacement for any existing internal system; it is an internship learning project.

2.2 User Classes and Characteristics

Role	Description
Admin	*Full system access; manages users, departments, and roles*
HR	*Manages employee records, leave approvals*
Manager	*Views team attendance, approves leave for their team*
Employee	*Views own profile, marks attendance, applies for leave*

2.3 Operating Environment

Backend: FastAPI (Python)
Frontend: React (served via browser, desktop-first)
Database: PostgreSQL

2.4 Design and Implementation Constraints

Backend must be built using FastAPI and Python
Frontend must be built using React
Relational database (PostgreSQL) is mandatory for structured HR data with relationships (employees, departments, leave)
Project deadline: Till the end of internship

2.5 Assumptions and Dependencies

Single organization/tenant use (no multi-company support)
Users are pre-registered by Admin/HR; no public self-signup
Internet connectivity assumed for all users

3. System Features

3.1 Authentication & Role-Based Access Control

Description: Allows users to securely log in and restricts access to features based on assigned role.
Priority: High.
Functional Requirements:

- FR-AUTH-01: System shall allow users to log in using email and password
- FR-AUTH-02: System shall issue a JWT token upon successful login
- FR-AUTH-03: System shall restrict API endpoints based on user role (Admin/HR/Manager/Employee)

- FR-AUTH-04: System shall reject invalid or expired tokens with appropriate error response

3.2 Employee Management

Description: Allows Admin/HR to manage employee records. Priority: High.
Functional Requirements:

- FR-EMP-01: Admin/HR shall be able to create a new employee record
- FR-EMP-02: Admin/HR shall be able to view, update, and deactivate employee records
- FR-EMP-03: Employees shall be able to view their own profile
- FR-EMP-04: Each employee shall be linked to a department

3.3 Department Management

Description: Allows Admin to manage organizational departments. Priority: Medium.
Functional Requirements:

- FR-DEPT-01: Admin shall be able to create, update, and delete departments
- FR-DEPT-02: System shall prevent deletion of a department with active employees assigned

3.4 Attendance Management

Description: Allows employees to mark daily attendance and allows Manager/HR to view attendance records. Priority: High.
Functional Requirements:

- FR-ATT-01: Employee shall be able to check in and check out once per day
- FR-ATT-02: System shall timestamp check-in/check-out automatically
- FR-ATT-03: Manager shall be able to view attendance of their team members
- FR-ATT-04: HR/Admin shall be able to view attendance across all employees

3.5 Leave Management

Description: Allows employees to request leave and managers/HR to approve or reject requests. Priority: High.
Functional Requirements:

- FR-LEAVE-01: Employee shall be able to submit a leave request with start date, end date, and reason
- FR-LEAVE-02: Manager shall be able to approve or reject leave requests for their team
- FR-LEAVE-03: System shall update leave balance upon approval
- FR-LEAVE-04: Employee shall be able to view status of submitted leave requests

3.6 Payroll Management

Description: Allows Admin/HR to generate and view monthly payroll records for employees based on basic salary and deductions. Priority: Medium.
Functional Requirements:

- FR-PAY-01: Admin/HR shall be able to generate a payroll record for an employee for a given month

- FR-PAY-02: System shall calculate net salary automatically as basic salary minus deductions
- FR-PAY-03: Employee shall be able to view their own payroll history
- FR-PAY-04: System shall prevent duplicate payroll generation for the same employee and month

4. Data Requirements

4.1 Logical Data Model

Core entities: User, Employee, Department,Payroll, Attendance, LeaveRequest, LeaveBalance, Role. Relationships: an Employee belongs to one Department; an Employee has many Attendance records; an Employee has many LeaveRequests; a LeaveRequest is approved by a Manager (User).

4.2 Reports

The system will support a basic attendance summary view (per employee, per month) and a leave history view.

4.3 Data Acquisition, Integrity, Retention, and Disposal

Employee records are soft-deleted (marked inactive) rather than permanently removed, to preserve historical attendance and leave data integrity.

5. External Interface Requirements

5.1 User Interfaces

The frontend will be a React-based web application with multiple views/pages including login, role-specific dashboards (Admin, HR, Manager, Employee), employee management screens, attendance tracking, and leave request/approval interfaces. Navigation between views will be handled client-side using React Router, providing a seamless, fast user experience without full page reloads.

5.2 Software Interfaces

Frontend (React) communicates with backend (FastAPI) via RESTful JSON APIs over HTTPS.

5.3 Communications Interfaces

Standard HTTPS REST API calls between client and server.

6. Quality Attributes

6.1 Usability

The interface shall be simple and role-appropriate, showing only relevant actions/data for each user role.

6.2 Performance

API responses shall return within 1-2 seconds under normal load for standard CRUD operations.

6.3 Security

Passwords shall be stored using secure hashing (bcrypt). All protected endpoints shall require valid JWT authentication. Role-based access control shall be enforced at the API level.

Appendix A: Glossary

- ***JWT*** *— JSON Web Token, used for stateless authentication*
- ***RBAC*** *— Role-Based Access Control*
- ***CRUD*** *— Create, Read, Update, Delete*

Appendix B: Analysis Models